

OASIS: Supplementary Materials

Author Names Redacted for Review

Affiliation Redacted
redacted@review.org

This document collects supplementary materials for the paper “OASIS: Feedback-Driven Marginal Statistics Correction for Cost-Based Query Optimizers.” It provides detailed scope and error-budget positioning (§ A), including the marginal-to-joint composition boundary, model architecture and instrumentation details (§ B), baseline fairness discussion and QuickSel-H adaptation (§ C), the PostgreSQL-style statistics-format adapter and conversion protocol (§ D), predicate-coverage sensitivity analysis (§ E), observation aggregation study details (§ F), PostgreSQL planner-only statistics-injection protocol details (§ G), OOD drift realism protocol (§ H), benchmark-inspired DML trace protocol (§ I), feedback-budget, feedback-noise, and deployment-safety diagnostic protocols (§ J), and detailed overhead breakdowns (§ K).

1 Scope and Error Budget

OASIS targets *single-column histogram staleness*. We explicitly exclude from scope:

- **Multi-column correlation errors:** Join cardinality misestimation caused by correlated column predicates requires multi-dimensional statistics (e.g., mvdist, DNF-based sketches, or learned joint models). OASIS operates on one-dimensional marginals and is complementary to multi-column methods.
- **Physical design errors:** Missing indexes, suboptimal partitioning, or incorrect materialized views are outside the statistics layer.
- **Cost-model calibration:** Even with correct selectivities, the optimizer’s cost model may mispredict operator runtimes. OASIS improves selectivity estimates but does not modify the cost model.
- **Data distribution shifts beyond drift:** Fundamental schema changes (column additions, type changes) or massive bulk loads that alter the distribution shape beyond what a compound drift operator captures are out of scope; these scenarios require a fresh **ANALYZE**.

Error budget decomposition. In a production optimizer, the total cardinality estimation error decomposes as:

$$\text{Total Error} = \underbrace{\text{Single-Column Error}}_{\text{OASIS target}} + \underbrace{\text{Correlation Error}}_{\text{multi-column methods}} + \underbrace{\text{Cost-Model Error}}_{\text{calibration}} + \underbrace{\text{Physical Design Error}}_{\text{indexing/partitioning}}.$$

OASIS addresses the first term. Even perfect single-column statistics do not eliminate the remaining terms, but accurate marginals are a necessary prerequisite: learned multi-column estimators (e.g., copula-based models like CoLSE [4])

combine one-dimensional marginals with a learned dependence structure, making single-column statistics quality a foundational concern.

Marginal-to-joint composition boundary. The multi-column composition experiments in the main paper should be interpreted as downstream use of corrected marginals, not as evidence that OASIS estimates correlations. The independence estimator uses the maximum-entropy joint distribution implied by the supplied marginals, while the Gaussian-copula estimator receives a fixed dependence structure. In this setting, plain OASIS marginals improve the joint estimator only modestly and inconsistently because they are not always feedback-consistent on the current observation window. OASIS-Proj is therefore a deployment calibration layer: it starts from the learned OASIS marginal correction, then applies a feedback-consistency projection before the marginal is passed to the downstream multi-column estimator. The core contribution remains single-column statistics correction.

2 Model and Instrumentation Details

This appendix collects implementation details trimmed from the main paper for page budget reasons.

2.1 Attention-Pooled MLP Architecture

The OASIS correction model is an *attention-pooled MLP* that exploits non-linear interactions between the prior histogram and the observation window. The architecture is deliberately lightweight for CPU deployment:

1. **Prior encoder:** The stale prior quantiles $\mathbf{v}_{\text{norm}}$ are first processed through a dedicated single-layer MLP ($9 \rightarrow 32 \xrightarrow{\text{ReLU}} 32$) to extract high-level distributional features before fusion with observations.
2. **Multi-head attention:** Each observation slot $o_i \in \mathbb{R}^{D_{\text{obs}}}$ receives $H=3$ independent attention scores $a_i^{(h)} = \mathbf{w}_{\text{attn}}^{(h)\top} o_i + b_{\text{attn}}^{(h)}$, one per head $h \in \{1, 2, 3\}$. Each head’s scores are masked (padded slots zeroed) and passed through softmax, yielding H attention weight vectors $\alpha^{(h)} \in \Delta^{K-1}$.
3. **Multi-head pooling:** Each head produces a pooled observation vector $\mathbf{p}^{(h)} = \sum_i \alpha_i^{(h)} o_i \in \mathbb{R}^{D_{\text{obs}}}$. The H pooled vectors are concatenated to form a multi-perspective observation summary $\mathbf{p}_{\text{all}} = [\mathbf{p}^{(1)} \parallel \dots \parallel \mathbf{p}^{(H)}]$.
4. **Deep residual MLP:** The context vector $\mathbf{c} = [\mathbf{v}_{\text{enc}} \parallel \mathbf{m} \parallel \mathbf{p}_{\text{all}}]$ (encoded prior $\parallel$ meta $\parallel$ multi-head pooled observations) is fed into a four-hidden-layer MLP with skip connections: $\mathbf{c} \rightarrow 128 \xrightarrow{\text{ReLU}} 128 \xrightarrow{\text{ReLU}} 64 \xrightarrow{\text{ReLU}} 64 \xrightarrow{\text{ReLU}}$ ($B-1$). Skip connections are added every two layers to facilitate gradient flow.
5. **Residual prediction:** The MLP output δ represents a *correction delta* rather than absolute quantile values. The final corrected histogram is computed as $\mathbf{v}_{\text{corrected}} = \mathbf{v}_{\text{norm}} + \delta$, allowing the model to focus on learning drift-induced deviations.

Training details. The OASIS MLP is trained offline on simulation-derived ground truth. A *compound drift operator* mutates an in-memory column through four atomic operations per round—**insert**, **delete**, **update**, and **null change**—with drift intensity q controlling the number of rounds between consecutive query observations. For each simulated sample the simulator records the true post-mutation quantile vector $\mathbf{v}^*$ as the regression target. The model is trained end-to-end via mini-batch Adam with He weight initialization, L2 regularization $\alpha=10^{-4}$, and gradient clipping (max norm = 1.0). The complete implementation is ~ 550 lines of pure Python + NumPy (no external ML framework), totalling ≈ 38 K parameters.

2.2 Per-Conjunct Operator Instrumentation

Filter and scan-filter-project operators are augmented with per-conjunct selectivity counters. During execution, whenever a batch of rows is processed through a compound predicate $\phi_1 \wedge \phi_2 \wedge \dots \wedge \phi_m$, each conjunct ϕ_j is evaluated independently on the full input batch:

$$n_j^{\text{in}} += |\text{batch}|, \quad n_j^{\text{out}} += |\{r \in \text{batch} : \phi_j(r)\}|.$$

The final intersection of all conjunct results produces the same output as the original compound filter while additionally recording per-conjunct independent selectivities $s_j^* = n_j^{\text{out}}/n_j^{\text{in}}$. Upon operator completion, these counters are written to the engine’s existing runtime metrics store. In the deployment model used by OASIS, the collector simply piggybacks on predicate evaluation already being performed by the engine, flushes a structured record to disk, and emits at most one observation tuple per filter conjunct. As a result, each query writes at most as many observation records as it has conjunctive filter clauses.

Algorithm 1 Per-Conjunct Operator Instrumentation

Require: Compound filter $\Phi = \phi_1 \wedge \dots \wedge \phi_m$, input batch B

Ensure: Filtered output B' , counters $n_j^{\text{in}}, n_j^{\text{out}}$

```

1: mask  $\leftarrow [\text{true}]^{|B|}$ 
2: for  $j = 1$  to  $m$  do
3:   result $_j \leftarrow \phi_j(B)$ 
4:    $n_j^{\text{in}} += |B|$ ;  $n_j^{\text{out}} += |\text{result}_j|$ 
5:   mask  $\leftarrow \text{mask} \wedge \text{result}_j$ 
6: end for
7: return  $B[\text{mask}]$ 

// On operator completion, emit per-conjunct metrics:
8: for  $j = 1$  to  $m$  do
9:   EMITMETRIC(conjunct_ $_j$ ,  $n_j^{\text{in}}$ ,  $n_j^{\text{out}}$ )
10: end for
```

2.3 QuickSel-H Adaptation Details

The QuickSel system [3] is a selectivity learner that fits a mixture-model density to satisfy feedback constraints. Because QuickSel outputs point selectivity estimates rather than histogram boundaries, we evaluate a clearly labeled **QuickSel-H** adaptation that bridges the gap between selectivity learning and histogram correction. The adaptation performs three steps:

1. **Mixture-model fitting:** Given the same stale prior H_0 and observation window $\mathcal{O}$ of size $K=16$, QuickSel-H fits a Gaussian mixture model with $M=5$ components whose CDF is constrained to match observed selectivities at the predicate values in $\mathcal{O}$. The optimization uses the same relative-entropy objective as the original QuickSel, solved via iterated scaling.
2. **Quantile extraction:** From the fitted mixture CDF, we extract the $B-1=9$ quantile values at the equi-depth levels used by all other methods. This ensures QuickSel-H outputs on the same histogram grid.
3. **Output interface:** The extracted quantiles are passed through the same monotonic projection as OASIS and returned as the corrected histogram.

Why this adaptation is necessary. QuickSel’s native output is a density estimate over the full domain, not a fixed-resolution histogram consumable by the CBO. To compare under a matched output interface, we must extract quantiles from this density. The extraction step introduces a minor information bottleneck (compressing a rich density into $B-1$ boundaries), but this is identical to the bottleneck all other methods face and does not fundamentally change QuickSel’s correction mechanism. We do not claim QuickSel-H is a line-by-line reproduction of the original system; it is a faithful adaptation that preserves QuickSel’s core density-fitting approach while conforming to the common output format required by our evaluation protocol.

3 Baseline Fairness Discussion

All feedback-driven methods compared in the main paper consume the same observation window of size $K=16$ and emit corrected quantiles on the same $B=10$ equi-depth histogram grid. No method receives additional information or computational budget beyond this shared interface.

STHoles. We implement the hierarchical refinement algorithm described in [1]. STHoles builds a tree of bucket refinements from feedback. We initialize it from the stale prior and allow up to K refinement steps per correction instance, using the same observation tuples provided to OASIS. The output quantiles are extracted from the refined tree on the standard $B-1$ grid.

ISOMER. We implement the consistency-based histogram adjustment from [2]. ISOMER adjusts bucket frequencies to satisfy observed selectivity constraints while minimizing KL divergence from the prior. We provide the same $K=16$ observation tuples and extract quantiles on the standard grid.

QuickSel-H. See §2 for the detailed QuickSel-H adaptation.

Simple heuristics. LinInterp and FeedAvg consume the same observation window but use fixed aggregation rules (blend factor 0.5 and exponential moving average, respectively). They are included to demonstrate that learned correction is necessary, not merely that any feedback-driven approach helps.

No method receives per-table or per-distribution tuning; all use default hyperparameters. Results are averaged over three independent training seeds (42, 123, 456) with 95% confidence intervals.

4 PostgreSQL-Style Statistics-Format Adapter

4.1 Motivation

OASIS corrects a canonical full-distribution view of a column. This is straightforward when a DBMS exposes a self-contained histogram, but many production systems do not. Instead, the optimizer’s statistics object often couples the histogram with auxiliary summaries whose probability mass must remain consistent after any update.

- **PostgreSQL:** Maintains a Most Common Values (MCV) list alongside a residual histogram that excludes MCV entries.
- **Oracle:** Uses top-frequency values together with height-balanced histograms.
- **SQL Server:** Employs a density vector together with histogram steps.

This coupling is the main adapter boundary for histogram-correction methods: *correcting the histogram alone* is often insufficient because the stored statistics object is not a standalone histogram. OASIS addresses this with a *statistics-format conversion interface* that translates a native statistics layout into a canonical full distribution, applies OASIS there, and translates the corrected result back.

We instantiate the interface on PostgreSQL’s MCV case because it is both practically important and representative of tightly coupled statistics architectures. The design isolates engine-specific adapter logic, but this paper validates only the PostgreSQL-style MCV+histogram instantiation.

4.2 Three-Phase Conversion Interface

The interface follows three phases:

1. **Reconstruction:** Convert engine-native statistics $\rightarrow$ canonical full distribution
2. **Correction:** Apply OASIS on the canonical representation
3. **Decomposition:** Convert corrected full distribution $\rightarrow$ engine-native statistics

This design decouples OASIS’s correction logic from PostgreSQL-specific storage fields. Other engines would require separate adapter implementations and validation.

4.3 PostgreSQL Instantiation

4.4 Problem Formulation

Given PostgreSQL statistics:

- MCV list: $\mathcal{M} = \{(v_1, f_1), (v_2, f_2), \dots, (v_k, f_k)\}$
- Residual histogram bounds: $\mathcal{H} = \{u_0, u_1, \dots, u_m\}$ where adjacent values define m equi-depth residual intervals
- Null fraction: f_{null}
- Non-null invariant: $\sum_{i=1}^k f_i + p_{\text{res}} = 1 - f_{\text{null}}$

Construct a canonical full distribution $\mathcal{F}$ that exposes the complete non-null column distribution to OASIS.

4.5 Phase 1: Reconstruction

Algorithm 2 Reconstruct Canonical Full Distribution from PostgreSQL Statistics

Require: MCV list $\mathcal{M}$, residual bounds $\mathcal{H} = \{u_0, \dots, u_m\}$, null fraction f_{null}

Ensure: Canonical full distribution $\mathcal{F} = (\mathcal{S}, \mathcal{R})$

- 1: Initialize $\mathcal{S} \leftarrow \emptyset, \mathcal{R} \leftarrow \emptyset$
 - 2: $p_{\text{res}} \leftarrow 1 - f_{\text{null}} - \sum_{(v_i, f_i) \in \mathcal{M}} f_i$
 - 3: **for** each $(v_i, f_i) \in \mathcal{M}$ **do**
 - 4: Add singleton bucket (v_i, f_i) to $\mathcal{S}$
 - 5: **end for**
 - 6: **for** $j = 1$ **to** m **do**
 - 7: Add range bucket $(u_{j-1}, u_j, p_{\text{res}}/m)$ to $\mathcal{R}$
 - 8: **end for**
 - 9: Validate: $\sum_{s \in \mathcal{S}} s.\text{freq} + \sum_{r \in \mathcal{R}} r.\text{freq} = 1 - f_{\text{null}}$ **return** $\mathcal{F} = (\mathcal{S}, \mathcal{R})$
-

PostgreSQL stores histogram *boundary values*, not explicit per-bucket frequencies. The conversion interface therefore interprets the residual histogram as m equal-depth intervals carrying equal shares of the residual non-MCV mass. Before tensorization, the interface evaluates the CDF of $\mathcal{F}$ at OASIS’s fixed quantile levels, yielding the canonical quantile vector consumed by the model.

4.6 Phase 2: OASIS Correction

Once reconstructed, the canonical full distribution is passed unchanged to the standard OASIS correction pipeline. The important systems property is that the model itself remains ignorant of the source database format; all engine-specific handling is localized to the conversion interface.

4.7 Phase 3: Decomposition

Given a corrected full distribution $\mathcal{F}' = (\mathcal{S}', \mathcal{R}')$, reconstruct PostgreSQL-compatible statistics:

- new MCV list: $\mathcal{M}'$
- new residual histogram bounds: $\mathcal{H}'$
- invariant: $\sum_i f'_i + p'_{\text{res}} = 1 - f_{\text{null}}$

Algorithm 3 Decompose Canonical Full Distribution to PostgreSQL Statistics

Require: Full distribution $\mathcal{F}' = (\mathcal{S}', \mathcal{R}')$, MCV threshold τ , MCV limit K , target histogram size $m+1$

Ensure: MCV list $\mathcal{M}'$, residual histogram bounds $\mathcal{H}'$

- 1: **Step 1: Extract MCV candidates**
 - 2: $\mathcal{C} \leftarrow \{(v, f) \mid (v, f) \in \mathcal{S}' \text{ and } f \geq \tau\}$
 - 3: Sort $\mathcal{C}$ by frequency (descending)
 - 4: $\mathcal{M}' \leftarrow$ top K entries from $\mathcal{C}$
 - 5: **Step 2: Build residual distribution**
 - 6: $V_{\text{mcv}} \leftarrow \{v \mid (v, f) \in \mathcal{M}'\}$
 - 7: Construct residual mixture from all range buckets in $\mathcal{R}'$ and all singletons in $\mathcal{S}'$ whose value is not in V_{mcv}
 - 8: **Step 3: Resample equi-depth histogram bounds**
 - 9: **for** $j = 0$ **to** m **do**
 - 10: Set u_j to the residual quantile at level j/m
 - 11: **end for**
 - 12: $\mathcal{H}' \leftarrow \{u_0, \dots, u_m\}$
 - 13: Validate: $\sum_{(v,f) \in \mathcal{M}'} f + p'_{\text{res}} = 1 - f_{\text{null}}$ **return** $\mathcal{M}', \mathcal{H}'$
-

4.8 MCV Selection Policy

The MCV threshold τ and limit K control the final decomposition:

- **High threshold:** keeps the MCV list compact but pushes more probability mass into the residual histogram
- **Low threshold:** retains more explicit singleton masses as MCV entries
- **PostgreSQL default-like regime:** $\tau \approx 0.01$, $K = 100$

This threshold is a decomposition policy choice, not part of OASIS’s core correction model. In our validation, experiments 1–3 remove threshold effects by using threshold-free round trips, while experiment 4 isolates the effect of τ_{MCV} after a simulated correction.

4.9 Adapter Boundary Beyond PostgreSQL

The PostgreSQL instantiation is only one instance of the conversion pattern. The same interface idea can be implemented for other engines that rely on histograms to very different degrees, but those implementations are outside the experimental scope of this paper:

- **Self-contained histograms** (MySQL, MariaDB): direct pass-through; conversion degenerates to a near-identity mapping.
- **MCV-coupled statistics** (PostgreSQL, Oracle): reconstruct from explicit frequent-value summaries plus residual histograms, then decompose back.

- **Hybrid summaries** (SQL Server): reconstruct from histogram steps plus density summaries, apply OASIS, then recompute auxiliary summaries.

The systems contribution here is to expose and validate this adapter boundary for a PostgreSQL-style statistics object, not to claim that all DBMS formats are already supported.

4.10 Consistency Guarantees

4.11 Probability Mass Conservation

Theorem 1 (Probability Conservation). *If the source statistics satisfy the non-null invariant $\sum_i f_i + p_{\text{res}} = 1 - f_{\text{null}}$, then:*

1. *reconstruction produces a canonical full distribution with the same non-null mass;*
2. *decomposition emits engine-native statistics whose MCV mass plus residual mass again equals $1 - f_{\text{null}}$.*

Proof. Reconstruction assigns all explicit MCV mass to singleton buckets and all remaining non-null mass to residual intervals by construction. Decomposition selects an explicit MCV set and resamples the remainder as residual histogram bounds, so the output MCV mass and residual mass partition the same corrected non-null mass. Therefore the invariant is preserved.

4.12 Approximate Idempotence

Definition 1 (Approximate Idempotence). *Let $(\mathcal{M}', \mathcal{H}') = \text{Decompose}(\text{Reconstruct}(\mathcal{M}, \mathcal{H}), \tau, K)$. The conversion interface is ϵ -idempotent if the round trip preserves total non-null mass within ϵ and preserves the residual quantile function within ϵ over a fixed evaluation grid.*

Perfect idempotence need not hold under thresholded decomposition because:

- the MCV threshold may change which values remain explicit,
- low-frequency singleton mass may be folded back into the residual distribution,
- floating-point rounding accumulates.

Our threshold-free validation shows exact total-mass preservation and machine-precision residual-quantile fidelity for the tested PostgreSQL-style statistics.

4.13 Implementation Interface

We provide a Python reference interface that exposes the conversion pattern explicitly:

Listing 1.1: Statistics-Format Conversion Interface (PostgreSQL Instance)

```

1 class PostgreSQLStatisticsAdapter:
2     def to_full_histogram(
3         self,
4         pg_stats: PostgreSQLStatistics
5     ) -> FullHistogram:
6         """Phase 1: Reconstruction"""
7         pass
8
9     def from_full_histogram(
10        self,
11        full_hist: FullHistogram
12    ) -> PostgreSQLStatistics:
13        """Phase 3: Decomposition"""
14        pass
15
16    def round_trip_test(
17        self,
18        pg_stats: PostgreSQLStatistics
19    ) -> Tuple[bool, str]:
20        """Validate round-trip fidelity"""
21        pass

```

The full reference interface is available in `appendix/mcv_adapter_interface.py`. This appendix is the supplementary counterpart of the compact PostgreSQL-style adapter discussion in the main text and is intended to be kept numerically consistent with `experiments/results/mcv_validation_results.json`.

4.14 Performance Considerations

4.15 Time Complexity

- **Reconstruction:** $O(k + n)$ where $k = |\mathcal{M}|$ and $n = |\mathcal{H}|$
- **Decomposition:** dominated by MCV candidate sorting and residual quantile sampling
- **Total overhead:** negligible relative to correction and well within planning-time constraints in our PostgreSQL validation

4.16 Space Complexity

The canonical full distribution is temporary and small:

$$\text{Space}(\mathcal{F}) = \text{Space}(\mathcal{M}) + \text{Space}(\mathcal{H}) + O(1).$$

In practice this overhead is acceptable because statistics objects are compact, the canonical distribution exists only during correction, and no persistent storage expansion is required.

4.17 Validation Summary

We validate the PostgreSQL instantiation of the conversion interface on synthetic PostgreSQL-style statistics spanning uniform, normal, Zipfian, and bimodal distributions.

Table 1: PostgreSQL Statistics-Format Interface Validation Summary

Metric	Result
Round-trip total mass error	0
Residual quantile MAE (mean)	2.4×10^{-17}
Residual quantile MAE (max)	9.2×10^{-17}
Selectivity MAE	1.1×10^{-4}
Selectivity P99	3.8×10^{-3}
Default-like overhead (100 MCV, 10–20 bins)	0.066–0.073 ms
Worst tested overhead (50 MCV, 50 bins)	1.03 ms

These results support three claims. First, the conversion interface is exact in mass preservation and essentially exact in residual-quantile reconstruction under threshold-free round trips. Second, it introduces negligible estimation bias and low latency on PostgreSQL-like settings. Third, threshold sensitivity is graceful: increasing τ_{MCV} shrinks the MCV list and gradually increases selectivity error, but does not compromise mass conservation.

4.18 Takeaway

The main value of this adapter is that it keeps histogram correction separate from native statistics storage details. OASIS corrects the canonical full-distribution view; the PostgreSQL-style adapter reconstructs and decomposes the native MCV+histogram object around that view. Extending the same pattern to other DBMSs is a design direction rather than an evaluated claim in this paper.

5 Predicate-Coverage Sensitivity Analysis

OASIS correction quality depends on the observation window covering the value range relevant to future predicates. We conduct a controlled sensitivity experiment at $q=10$ by restricting observations to a “hot region” of width w centered at the domain midpoint.

Table 2: Predicate-Coverage Sensitivity at $q=10$

Coverage w	Stale Prior Q-Error	OASIS Q-Error	Improvement
1.00 (full)	2.720	1.232	55%
0.70	2.720	1.268	53%
0.50	2.720	1.395	49%
0.30	2.720	1.625	40%
0.15	2.720	1.890	30%

At $w=0.7$ (70% coverage), Q-Error degrades by only 3% relative to full coverage. At $w=0.3$, degradation reaches 40% but OASIS still outperforms the stale

prior by 26%. Even at $w=0.15$ (15% coverage), OASIS provides 30% improvement over the stale prior, demonstrating graceful degradation under sparse feedback. The abstention policy (§H in the main paper) provides a natural guard for extreme sparsity scenarios.

6 Observation Aggregation Study

The main paper reports that replacing attention pooling with mean/max pooling degrades Q-Error by 12%/18% at $q=10$. Here we provide the full breakdown.

Table 3: Observation Aggregation Ablation at $q=10$

Pooling Strategy	Q-Error Degradation vs. Attention	
Attention (OASIS)	1.232	—
Mean pooling	1.380	+12%
Max pooling	1.453	+18%

Mean pooling treats all observations equally, diluting the signal from highly informative predicates. Max pooling amplifies the most extreme observation but is sensitive to outliers. Attention learns non-uniform weights that up-weight informative observations while down-weighting redundant or noisy ones. This advantage is most pronounced when the observation window contains heterogeneous predicates of varying selectivity and informativeness.

7 PostgreSQL Planner-Only Statistics-Injection Protocol

The planner-only evaluation protocol uses PostgreSQL’s `EXPLAIN` (without `ANALYZE`) to validate the statistics→estimated-cardinality→plan-shape path. `EXPLAIN` invokes PostgreSQL’s optimizer and cost model, but it does not execute the query. The protocol therefore provides optimizer-facing evidence without making wall-clock runtime claims.

Protocol steps.

1. Create a synthetic PostgreSQL schema with a fact table and dimension table. The fact table stores the current post-drift data.
2. Generate stale and fresh single-column distributions for `fact.x`; build ISOMER, OASIS, OASIS-Proj, and Hybrid corrected marginal statistics from the same feedback window.
3. For each statistics state, create an analyzed one-column source table and copy its typed `pg_statistic` row into the target `fact.x` statistics entry. This avoids hand-writing PostgreSQL’s `anyarray` catalog fields.
4. For each test query, obtain true cardinality using `COUNT(*)` and obtain PostgreSQL estimated rows and plan shape using `EXPLAIN (FORMAT JSON)`.
5. Compare row-estimation Q-error, plan-shape match to the fresh-statistics planner, recovery of stale/fresh plan disagreements, and new deviations from fresh when stale already matched fresh.

Table 4: Representative PostgreSQL plan changes caused by corrected statistics. Each row is a query where stale statistics disagree with fresh statistics and OASIS-Proj matches the fresh plan shape.

Family	Predicate	True	Method	Est. rows	Row QE	Root	Plan summary
Join + dimension filter	le_0.045	14	Stale	8292	592.3	Nested Loop	Nested Loop: Nested Loop:Inner; Bitmap Heap Scan:dim; Bitmap Index Scan:dim_payload_idx...
Join + dimension filter	le_0.045	14	OASIS-Proj	686	49.0	Hash Join	Hash Join: Hash Join:Inner; Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx; Bitmap...
Join + dimension filter	le_0.045	14	Fresh	95	6.8	Hash Join	Hash Join: Hash Join:Inner; Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx; Bitmap...
Selection	le_0.045	128	Stale	55041	430.0	Seq Scan	Seq Scan: Seq Scan:fact
Selection	le_0.045	128	OASIS-Proj	4543	35.5	Bitmap Heap Scan	Bitmap Heap Scan: Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx
Selection	le_0.045	128	Fresh	590	4.6	Bitmap Heap Scan	Bitmap Heap Scan: Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx
Join	bt_0.13_0.2	14073	Stale	144	97.7	Nested Loop	Nested Loop: Nested Loop:Inner; Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx; In...
Join	bt_0.13_0.2	14073	OASIS-Proj	13886	1.0	Hash Join	Hash Join: Hash Join:Inner; Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx; Seq Sc...
Join	bt_0.13_0.2	14073	Fresh	14023	1.0	Hash Join	Hash Join: Hash Join:Inner; Bitmap Heap Scan:fact; Bitmap Index Scan:fact_x_idx; Seq Sc...

Results. The formal batch covers 12 configurations: two drift directions (**left_shift** and **right_shift**), two fact-table sizes (100K and 200K rows), and three random seeds. Each configuration evaluates 84 scan/join query templates under each statistics state. No query runtime is measured.

- Stale statistics produce aggregate row Q-error 27.748 and match the fresh-statistics plan shape on 56.0% of queries.
- Plain OASIS reduces row Q-error to 3.306 and recovers 91.7% of stale/fresh plan-shape disagreements, but introduces new plan deviations in 13.8% of cases where stale statistics already matched fresh.
- OASIS-Proj reduces row Q-error to 2.360, matches the fresh-statistics plan shape on 96.1% of queries, recovers 92.6% of stale/fresh plan disagreements, and reduces new plan deviations to 1.1%.
- Across configurations, OASIS-Proj has mean row Q-error 2.375 ± 0.275 , close to ISOMER’s 2.372 ± 0.283 and below plain OASIS’s 3.313 ± 0.228 .

This protocol validates the causal chain statistics correction $\rightarrow$ estimated rows $\rightarrow$ cost re-estimation $\rightarrow$ plan shape inside a real PostgreSQL optimizer, while intentionally avoiding runtime claims that would depend on executor, cache, and system effects outside the single-column statistics layer.

Table 4 gives representative plan-shape changes from the batch. These examples are not used as runtime evidence. They show the mechanism that the planner-only experiment is intended to validate: stale statistics can move estimated rows enough to select a different root operator or join strategy, while OASIS-Proj moves the estimated rows and plan shape back toward the fresh-statistics planner.

8 OOD Drift Realism Protocol

The OOD drift realism suite evaluates the same OASIS checkpoint used in the main synthetic experiments. The checkpoint is trained only on the compound drift generator; no samples from the OOD families are used for retraining or hyperparameter tuning. Each OOD case follows the same causal protocol as the main synthetic suite: capture stale quantiles, interleave drift steps with 16 feedback observations, compute final post-drift quantiles, and evaluate future selectivity estimates against the final distribution.

The six OOD families are deployment-inspired patterns implemented in `extended_drift_generators.py`: batch loading, range shift, skew evolution, outlier burst,

Table 5: Benchmark-inspired DML trace sanity check. Each trace emits explicit insert/update/delete events anchored to an analytical benchmark table/column pattern. Values are geometric mean selectivity Q-error; no production trace or query runtime is used.

Trace	DML mix I/U/D	Growth	Stale	ISOMER	OASIS	OASIS-Proj	Hybrid	Fresh
Sales append	96/0/4	+79.9%	1.529	1.128	1.132	1.121	1.135	1.000
Promotion price	30/70/0	+33.9%	1.025	1.018	1.086	1.026	1.019	1.000
Inventory restock	19/75/6	+19.8%	1.222	1.042	1.171	1.064	1.042	1.000
Returns/cancel	38/17/45	-3.6%	2.047	1.163	1.219	1.142	1.162	1.000
Customer churn	36/62/2	+41.2%	1.172	1.057	1.138	1.076	1.061	1.000
Seasonal mixed	58/38/4	+50.8%	1.029	1.020	1.107	1.031	1.021	1.000

multimodal drift, and seasonal/mixed drift. The reported table uses 128 held-out cases per family, $B = 10$ histogram buckets, and 64 future CDF probe points per case. This experiment is not intended to prove that the simulator is a production DML trace; it tests whether OASIS and OASIS-Proj remain useful when the drift mechanism differs from the training generator.

9 Benchmark-Inspired DML Trace Protocol

The trace-grounded sanity check is not a production-log experiment. It constructs explicit DML event streams whose table/column stories are anchored to analytical benchmark maintenance patterns: append-heavy sales dates, promotion price revisions, inventory restocking, returns and cancellations, customer segment churn, and seasonal mixed maintenance. Values are normalized to the unit interval so that the same single-column correction interface is used as in the synthetic and OOD experiments.

Each trace case begins with a stale histogram, then executes 16 DML steps. Each step emits concrete insert, update, and delete counts and then records one predicate feedback observation against the current post-step table. The reported run uses 96 cases per trace, $B = 10$ histogram buckets, and 64 future CDF probe points per case. The same OASIS checkpoint trained on compound drift is used unchanged; no trace-specific retraining or hyperparameter tuning is performed.

The table should be interpreted as a realism sanity check rather than a claim that the traces match a particular deployment. Its main role is to expose a useful boundary: OASIS-Proj and Hybrid consistently improve over stale statistics, while plain OASIS can regress on mild update-heavy traces where feedback consistency is more valuable than learned global extrapolation.

10 Feedback Budget, Feedback Noise, and Safety Diagnostics

The feedback-budget experiment truncates each held-out sample to the K most recent observations, with $K \in \{2, 4, 8, 16\}$. The trained OASIS checkpoint is

not retrained for smaller budgets; instead, the truncated window is padded to the checkpoint’s native 16-slot tensor width. ISOMER, OASIS-Proj, and Hybrid receive the same truncated observation window, so the comparison isolates the effect of feedback availability.

The feedback-noise experiment perturbs only the observed actual selectivities in the feedback window. It applies multiplicative log-normal noise with $\sigma \in \{0, 0.02, 0.05, 0.10\}$, keeps stale priors and fresh future selectivities fixed, and evaluates the same generator-driven optimizer-decision proxy used in the main paper. The reported diagnostic uses 64 held-out cases per drift intensity and one noise seed, because noisy projection constraints are more expensive than the clean-feedback setting.

Both diagnostics are optimizer-facing rather than runtime-facing. The scan and join decisions are selected from estimated selectivities under a fixed cost proxy, then scored by true proxy cost using the generated post-drift distribution. The purpose is to characterize deployment sensitivity, not to claim wall-clock speedups.

The deployment-safety table in the main paper is derived from cached outputs of four diagnostics: PostgreSQL planner-only batch summaries, feedback-budget sensitivity, feedback-noise robustness, and the benchmark-inspired DML trace sanity check. It does not introduce an oracle selector. Projection uses only the current feedback window, and Hybrid chooses among stale, ISOMER, OASIS, and OASIS-Proj by minimizing feedback-window residual. The table is generated by `experiments/deployment_safety_analysis.py`.

11 Stage-2 Calibration: Soft Projection, Banded Projection, and the Calibrated Router

This section gives the protocol and full per-family results behind the main paper’s finding that the Stage-2 calibration is regime-dependent.

11.1 Operators

All Stage-2 operators act on the same interval partition induced by the active feedback predicates and start from the OASIS-noProj histogram. *Hard projection* (the default OASIS Stage 2) performs cyclic I-projections so that every active predicate’s interval mass matches its observed selectivity exactly, dropping the oldest constraints when the active set becomes infeasible. *Soft projection* instead minimizes $\text{KL}(p \| p_{\text{noProj}}) + \lambda \sum_j w_j (A_j p - y_j)^2$ by mirror descent over cell masses, where A_j is the interval mask of feedback j and y_j its observed selectivity. We study three weightings: full-window ($w_j=1$), a fixed recent window ($w_j=1$ for the last $k=8$ observations, 0 otherwise), and *conflict-aware*. The conflict-aware weighting fits a hard reference distribution p_{ref} to the most recent k observations, scores each constraint by $\text{conflict}_j = |A_j p_{\text{ref}} - y_j|$, and sets $w_j = \exp(-(\text{conflict}_j/\tau)^2)$ with $\tau=0.03$, so feedback

contradicted by the recent observations is suppressed while old-but-consistent feedback is retained. *Banded projection* I-projects p_{noProj} onto the bands $y_j \pm \kappa\sqrt{y_j(1-y_j)}$ ($\kappa=0$ recovers hard projection). The *calibrated router* selects, per case, the candidate with the lowest feedback-window residual among {stale, ISOMER, OASIS-noProj, hard projection, conflict-aware soft}. All operators and the diagnostic are implemented in `cdf_kll_ml_pipeline/modern_baselines.py` (`correct_soft_isomer`, `correct_band_isomer`) and `experiments/oasis_soft_projection_diagnostics.py`.

11.2 Evaluation Protocol

The single-column comparison reuses the cached compound-drift data and the fixed OASIS checkpoint; it does not retrain. For each case it generates held-out future predicates, estimates their selectivity from each Stage-2 output, and reports the geometric-mean Q-error, together with the feedback residual on the observation window, quantile MAE against fresh statistics, and an optimizer-proxy join regret. This future-predicate aggregation differs from the `single_column_projection` table in the main results (which reports a different per-case aggregation), so the numbers here are not directly comparable to that table and are presented as a self-contained operator comparison. The downstream and safety suites (OOD drift, DML trace, FactorJoin, composition family, optimizer proxy, PostgreSQL planner subset) reuse the same protocols documented in the preceding sections, with the soft/router methods added.

11.3 Single-Column and Optimizer-Proxy Results

Table 6: Stage-2 calibration variants on held-out future predicates (expanded cached drift, fixed checkpoint) and on the optimizer-decision proxy. Lower is better.

Stage-2 variant	Single-column				Optimizer proxy	
	Sel. QErr	Resid.	Q. MAE	JoinReg	Sel. QErr	JoinReg
Hard projection (OASIS-Proj)	1.442	0.0440	0.0519	1.0344	1.446	1.0348
Full-window soft	1.376	0.0348	0.0483	1.0284	1.378	1.0285
Recent-window soft ($k=8$)	1.408	0.0404	0.0471	1.0322	1.408	1.0318
Conflict-aware soft	1.401	0.0392	0.0473	1.0313	1.404	1.0312
Hybrid (no soft)	1.429	0.0395	0.0513	1.0326	1.430	1.0326
Calibrated router	1.395	0.0358	0.0486	1.0302	1.396	1.0297

Table 6 shows that full-window soft has the best raw single-column accuracy but, as the safety suites below show, is unsafe; the calibrated router recovers most of that accuracy (1.395) while remaining deployment-safe, and has the lowest feedback residual, join regret, and (not shown) new-risk loss (1.37% versus hard 1.81%). The router routes to the conflict-aware soft candidate on 44% of cases and to hard/ISOMER/OASIS-noProj otherwise. As a negative control, banded

projection monotonically *worsens* single-column Q-error as the band widens: $\kappa \in \{0.04, 0.10, 0.20, 0.35\}$ gives Q-error $\{1.437, 1.450, 1.470, 1.509\}$, confirming that the gain comes from per-case routing, not from relaxing the feedback constraints.

11.4 Out-of-Distribution and Trace Safety

Table 7: Future selectivity Q-error by OOD drift family and DML trace. Soft columns use no recency decay; conflict-aware uses $\tau=0.03$. Full-window soft fails on the strong batch-load and range-shift families and on append/returns traces; the recent-window and conflict-aware variants remove those failures.

Family / trace	Hard (Proj)	Recent8 soft	Conflict soft	Full-window soft
OOD batch load	1.272	1.271	1.336	1.482
OOD range shift	1.187	1.177	1.235	1.284
OOD skew evolution	1.171	1.174	1.169	1.174
OOD outlier burst	1.055	1.055	1.056	1.061
OOD multimodal	1.060	1.068	1.066	1.064
OOD seasonal/mixed	1.057	1.056	1.059	1.066
Trace sales append	1.121	—	1.135	1.190
Trace returns/cancel	1.142	—	1.158	1.204

Table 7 documents why soft projection needs a consistency gate. Full-window soft softly fits stale, contradicted feedback and fails on abrupt drift (batch load 1.482, range shift 1.284) and on visibly shifted traces (sales append 1.190, returns 1.204). The conflict-aware variant removes the trace and milder-OOD failures and is best on skew evolution, but it still trails hard/recent8 on the two strongest OOD families: some pre-drift observations are coincidentally consistent with the recent reference ($\text{conflict}_j \approx 0$) and survive any τ , whereas the recent-window age cutoff removes them unconditionally. Reducing τ to 0.015 leaves single-column accuracy unchanged (1.403) and improves batch/range only marginally (1.322 / 1.232), confirming this is an inherent property of consistency-based gating rather than a tuning artifact. The calibrated router sidesteps the issue: it routes to hard/ISOMER on these workloads (the soft candidate has higher residual), so router safety equals the deployed Hybrid.

11.5 Downstream Estimators and PostgreSQL Plan-Shape

On the FactorJoin join kernel, conflict-aware soft alone gives all-drift join Q-error 1.0367 (recent-window soft 1.038), still behind hard projection 1.024 and the residual-gated router 1.0174, which never selects soft because soft’s join feedback residual (0.0153) exceeds hard projection’s (0.0070). On the six-estimator composition family, conflict-aware soft is within 1–2% of hard projection on every estimator (independence 1.222 vs. 1.191; IPF/Sinkhorn 1.284 vs. 1.280; Gaussian 1.239 vs. 1.224; Clayton 1.273 vs. 1.256; Gumbel 1.245 vs. 1.234; Frank 1.274 vs. 1.260), so the router again prefers hard/Hybrid.

Table 8: PostgreSQL planner-only six-configuration subset (left/right/bimodal shift $\times$ two seeds, 504 queries). The calibrated router is identical to the Hybrid because the residual gate routes to ISOMER and never selects soft.

Method	Row QErr	Fresh-plan match	New plan deviations
Hard projection (OASIS-Proj)	2.237	90.9%	6.86%
Conflict-aware soft	2.277	91.1%	6.54%
Hybrid (no soft)	2.234	90.1%	8.17%
Calibrated router	2.234	90.1%	8.17%

Table 8 confirms that adding the soft candidate to the router introduces no PostgreSQL plan-shape regression: the calibrated router is byte-identical to the Hybrid across all 504 queries. The residual gate’s preference for ISOMER over hard projection on these planner-facing cases (which is why both the Hybrid and the router show 8.17% new deviations versus hard projection’s 6.86%) is a property of the residual criterion, not of the soft candidate; a plan-shape-aware gate is left to future work.

12 Detailed Overhead Breakdown

Table 9: OASIS Runtime Overhead Breakdown (1000 trials)

Component	p50 (ms)	p95 (ms)	p99 (ms)
Feature tensorization	0.38	0.42	0.45
Model forward pass	0.66	0.72	0.78
Total correction	1.04	1.18	1.25
Statistics-format conversion	0.07	0.07	0.07
Combined correction + conversion	1.11	1.25	1.32
Feedback collection (per tuple batch)	0.12	0.18	0.23

For comparison, a column-level **ANALYZE** on a 10^7 -row table costs an estimated 300–500 ms of sequential I/O. OASIS’s combined latency of ≈ 1.13 ms is $\approx 300\times$ cheaper per invocation, well within the 200 ms planning budget (desideratum **D2**).

The statistics-format conversion interface adds negligible overhead (0.066–0.073 ms with PostgreSQL-like defaults of 100 MCV entries and 10–20 residual bins). Even in the worst tested configuration (50 MCV, 50 bins), the conversion completes in 1.03 ms.

References

1. Bruno, N., Chaudhuri, S., Gravano, L.: STHoles: A multidimensional workload-aware histogram. In: Proceedings of SIGMOD. pp. 211–222 (2001)

2. Markl, V., Megiddo, N., Kutsch, M., Tran, T.M., Lang, P., Raman, V.: Consistent selectivity estimation via maximum entropy. *The VLDB Journal* **16**(2), 169–188 (2007)
3. Park, Y., Zhong, S., Mozafari, B.: QuickSel: Quick selectivity learning with mixture models. In: *Proceedings of SIGMOD*. pp. 1017–1033 (2020)
4. Rathuwadu, L., Liu, G., Leckie, C., Borovica-Gajic, R.: Colse: A lightweight and robust hybrid learned model for single-table cardinality estimation using joint cdf. *arXiv preprint arXiv:2512.12624* (2025). <https://doi.org/10.48550/arXiv.2512.12624>